

Bengali LLM Quality Evaluation Study

Bangladesh Locale — Final Evaluation Report

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Role: Independent Human Evaluator — Bengali LLM Quality

Locale: Bangladesh

Evaluation Date: 1 October 2026

50-case independent human evaluation | Gemini Web — “3.6 Flash” | 1 October 2026

This report summarizes an independent, self-directed human evaluation of 50 Bengali LLM outputs designed to assess quality in a Bangladesh-language and cultural context. The study was conducted as a portfolio-based evaluation exercise using a structured quality assessment framework.

1. Executive Summary

The evaluation covered 50 controlled Bengali prompts across 10 categories, with five cases per category. 44 cases (88%) were classified as Pass and 6 (12%) as Fail. Six cases contained material enough issues to receive a Fail classification; two were Major and four were Minor. Four additional cases contained documented Minor issues but were retained as Pass because the core task outcome remained correct.

The strongest results were observed in Grammar, Spelling, Factuality, Instruction Following, Meaning Preservation, and Bangladesh Localization, where all five cases in each category passed. The most frequent material weaknesses appeared in Tone/Register, followed by Bangladesh-vs-West-Bengal cultural comparison, Translation Fidelity, and Formatting/Terminology.

2. Study Scope and Methodology

Parameter	Study setting
Language	Bangla
Locale	Bangladesh
Model	Gemini Web — model label “3.6 Flash”
Evaluation date	1 October 2026
Sample	50 controlled prompts
Categories	10 categories × 5 prompts
Evaluator	IMH
Evaluation approach	Human review against task-specific quality criteria and a documented error/severity framework

The evaluation distinguishes source-text problems from model-output problems. Multiple acceptable Bengali formulations were not treated as errors merely because they differed stylistically. Extra relevant information was not penalized when the prompt did not prohibit it. A Fail classification was reserved for an issue that materially affected the requested task; Minor issues could be documented without overturning a substantively correct response.

3. Overall Statistics

Metric	Count	Share
Total cases	50	100%
Pass	44	88%
Fail	6	12%
Major severity	2	4%
Minor severity	8	16%
Cases with documented issue	10	20%

Note: severity is populated only for cases where an issue was explicitly documented in the dataset. Therefore, the 8 Minor and 2 Major counts should not be interpreted as a severity rating for every case.

4. Category-wise Results

Category	Cases	Pass	Fail	Fail rate
Grammar	5	5	0	0%
Spelling	5	5	0	0%
Bangladesh Localization	5	5	0	0%
Bangladesh vs West Bengal	5	4	1	20%
Translation Fidelity	5	4	1	20%
Meaning Preservation	5	5	0	0%
Factuality	5	5	0	0%
Instruction Following	5	5	0	0%
Tone/Register	5	2	3	60%
Formatting/Terminology	5	4	1	20%

Category-level results show a clear concentration of failures rather than a uniform quality problem. Six categories recorded no Fail cases, while Tone/Register recorded 3 Fail cases out of 5.

5. Material Error Cases

BQA-018 — Bangladesh vs West Bengal — Others — Major

Model correctly identifies the shared agricultural/new-harvest basis of Nabanna, but it makes an overly broad cultural contrast between Bangladesh and West Bengal. It characterizes Bangladesh as uniformly secular/national and West Bengal as primarily Hindu religious/family-centered. Evidence from West Bengal shows Nabanna is also practiced as a local agrarian folk tradition, while religious offerings can be part of some local practices rather than defining the whole region. The answer therefore risks presenting regional cultural diversity as a binary distinction.

BQA-022 — Translation Fidelity — TRANSLATION-FIDELITY — Major

The primary translation, “শুক্রবার ব্যাপ্তির মধ্যে প্রতিবেদনটি জমা দিন,” is unnatural and does not faithfully convey the intended Bangla expression for “by Friday.” The alternative “অনুগ্রহ করে শুক্রবারের মধ্যে রিপোর্টটি জমা দিন।” is natural and preserves the original polite imperative tone. However, presenting the incorrect primary translation is a material translation-quality issue.

BQA-041 — Tone/Register — FLUENCY — Minor

The response is generally polite, formal, and appropriate for communication with a government official. However, the phrase “পর্যালোচনা ও পরবর্তী পর্যালোচনার জন্য” is redundant and awkward, as the same concept is unnecessarily repeated.

BQA-043 — Tone/Register — MEANING-PRESERVATION — Minor

The response maintains a respectful and professional customer-support tone and avoids making an absolute resolution promise. However, the sentence “বিষয়টি আমাদের নজরে আনার জন্য আমরা আন্তরিকভাবে দুঃখ প্রকাশ করছি” is contextually inappropriate because bringing the issue to the company's attention is not itself the cause for regret. The response should thank the customer for bringing the issue to their attention and express regret separately for the inconvenience caused.

BQA-045 — Tone/Register — TONE-REGISTER — Minor

The response is concise and professionally structured, but “উপস্থিতি নিশ্চিত করা আবশ্যিক” makes the instruction sound mandatory rather than a polite request to attend on time. Since the prompt specifically asks to “অনুরোধ করা হবে”, a softer formulation such as “সময়মতো উপস্থিত থাকার জন্য অনুরোধ করা হলো” would better match the requested tone.

BQA-047 — Formatting/Terminology — TERMINOLOGY — Minor

The response correctly provides Bengali equivalents for all three English terms and keeps the English terms beside them. However, for “machine learning” it proposes both “যন্ত্র শিক্ষা” and “মেশিন লার্নিং”, despite the instruction asking for one consistent Bengali term for each item. The alternatives make the terminology less consistent.

6. Key Findings

- Instruction compliance was strong: all five Instruction Following cases passed.
- Basic Bengali quality was strong in the sampled Grammar and Spelling cases; all five in each category passed.
- Bangladesh-specific localization was successful in the five sampled cases, although some answers contained overgeneralized practical requirements.
- Cultural comparison requires more caution: BQA-018 was failed because a nuanced Nabanna comparison was presented as an overly broad Bangladesh-versus-West-Bengal binary.
- Translation quality was generally strong, but BQA-022 demonstrated that an unnatural primary translation can be a material failure even when a good alternative is also supplied.
- Tone/Register was the most failure-prone category in this sample, with issues involving redundancy, contextual apology wording, and excessive formality/mandatory wording.
- Terminology consistency can matter even when each individual term is understandable: BQA-047 offered two Bengali options for one requested term, reducing consistency.

7. Representative QA Patterns

Case	QA dimension	Observed issue
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BQA-018	Cultural localization	Overgeneralized regional-cultural contrast; Bangladesh and West Bengal practices were treated too categorically.
BQA-022	Translation fidelity	Primary translation was unnatural for “by Friday,” despite a correct alternative.
BQA-041	Formal Bengali fluency	Redundant phrase: “পর্যালোচনা ও পরবর্তী পর্যালোচনার জন্য.”
BQA-043	Meaning/context	Apology was attached to the customer’s act of reporting the issue rather than to the inconvenience.
BQA-045	Tone/register	“আবশ্যিক” shifted a requested polite request toward a mandatory instruction.
BQA-047	Terminology consistency	Two Bengali renderings were offered for “machine learning” although one was requested.

8. Limitations

- The sample contains 50 controlled prompts and should not be interpreted as a statistically representative estimate of all Bengali LLM performance.
- The study evaluates one model label/version and one evaluation date; model behavior may change over time.
- Some prompts permit multiple valid Bengali formulations, so human judgment remains relevant in fluency, tone, and terminology decisions.
- The dataset is designed as a portfolio-quality demonstration rather than a commissioned benchmark or independent scientific study.
- Where practical Bangladesh-specific requirements were discussed, the evaluation focused on whether claims were appropriately scoped; it did not attempt to establish a universal admission or employment standard.

9. Reproducibility and Evaluation Record

Evaluated using the Gemini web interface with model label “3.6 Flash” on 2026-10-01. The accompanying Excel dataset contains the 50 prompts, raw outputs, error classifications, severity, evaluation notes, corrections where applicable, and evaluator initials.

10. Conclusion

Within this 50-case Bangladesh-locale sample, the evaluated Bengali outputs were predominantly successful, with an 88% Pass rate. The failures were concentrated in higher-context tasks rather than basic grammar or instruction-following: cultural generalization, translation naturalness, contextual meaning, register, and terminology consistency. These results illustrate why Bengali LLM quality assurance requires more than surface grammar checking; evaluators also need to assess meaning, local context, register, and instruction fidelity.

Prepared for portfolio use — Bengali LLM Quality Evaluation Study